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OPTICAL SENSING DEVICE AND METHOD FOR MANUFACTURING THE SAME

Abstract

A light-sensing device and a manufacturing method thereof are provided. The light-sensing device includes a substrate, at least one light-emitting component and a light-sensing component. The light-emitting component is used to emit a light, and the light-sensing component is used to receive the diffuse-reflected light after the light is diffuse-reflected by an external object. Moreover, the light-emitting component and the light-sensing component are individually and separately mounted on the substrate.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

[0001] This application claims priority to Taiwan Patent Application No. 113107849 filed on Mar. 5, 2024, which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to a light-sensing device and a method for manufacturing the same, in particular, a light-sensing device with multiple combinations of flexibility and measurement accuracy and a manufacturing method thereof.

Descriptions of the Related Art

[0003] Non-invasive blood glucose testing is a technology that measures blood glucose levels without puncturing the skin or taking a blood sample. Such technology has important application values in reducing the inconvenience of diabetic patients and improving the frequency and convenience of blood glucose monitoring. Near-infrared spectroscopy (NIRS) blood glucose detection technology is a non-invasive optical blood glucose detection technology which utilize near-infrared light with a wavelength ranges from 700 nanometers (nm) to 2500 nanometers (nm). The infrared light in this wavelength range can be absorbed and scattered by components such as water, fat, and protein in tissues. This technology is used to presume the blood sugar concentration in human tissues by measuring the changes in absorption, reflection and scattering of human tissues before and after infrared light is irradiated to human tissues.

[0004] As shown in FIG. 1, which shows a schematic cross-sectional view of a conventional near-infrared blood glucose detection device 1. A conventional blood glucose detection device is usually a multi-chip module 10. When manufacturing a multi-chip module, several light-sensing components and several light-emitting components are solidified onto a common conductive substrate. After that, several multi-chip modules are formed through processes such as wire bonding, film pasting, light-blocking component production, and cutting. Then, the multi-chip module 10 is mounted on a printed circuit board 20 to finish the whole process. Each multi-chip module 10 includes a light-sensing component 11 and a plurality of light-emitting components 12 which are bonded on a conductive substrate 13. This is a design of a light-blocking component 14 arranged between each light-sensing component 11 and the light-emitting components 12.

[0005] The light-blocking component 14 of the conventional blood glucose detection device 1 shown in FIG. 1 is designed as a half-cut and glue-dispensing structure. After the above-mentioned multi-chip module 10 has been processed by the packaging process of the protective layer 15 and the optical film cover 16, and then cuts the package body between the light-emitting component and the light-sensing component in a half-cutting manner to form a trench for the intended light-blocking component. The grooves are then filled with glue to form the light-blocking component 14. One of the shortcomings of the conventional blood glucose detection device 1 is that the light-blocking component 14 only exists between the light-sensing component 11 and the light-emitting component 12 and can only block the light transmission between the light-sensing component and the light-emitting component. There is no light-blocking component design around the multi-chip module 10 so it cannot block the entry of ambient light from the outside, and it cannot prevent the light-emitting components from leaking to unintended external environments. Moreover, the above-mentioned conventional half-cutting and dispensing processes also have shortcomings such as complicated process, low yield, slow production, and high cost.

[0006] In addition, when manufacturing a multi-chip module, several light-sensing components and several light-emitting components are bonded on a common conductive substrate. The configuration of each chip in the multi-chip module formed after wire bonding, cutting and other

processes has been fixed. At this time, if any chip is detected to be defective during the manufacturing process, any chip in the multi-chip module cannot be replaced. The entire module can only be discarded and there is no possibility of replacement.

[0007] Please refer to FIG. 2A and FIG. 2B together, which shows that the multi-chip module **10** in the conventional blood glucose detection device may be worn too loosely or too tightly due to the difference in the user's operation status and the user's human tissue. As shown in FIG. 2A, if the gap between the blood glucose detection device and human tissue is too loose, the multi-chip module **10** may suffer from optical signal loss. On the other hand, as shown in FIG. 2B, if the gap between the blood glucose detection device and the human tissue is too tight, the multi-chip module **10** will excessively press the human tissue and will force the glucose **30** in the human tissue to be expelled outward to other locations inside the human tissue. Accordingly, the blood glucose concentration near the measured area will decrease, and indirectly, the measurement results will be distorted and measurement accuracy will be affected. In order to overcome the above problems, the industry urgently needs an innovative blood glucose detection device and its manufacturing method to improve the above problems.

SUMMARY OF THE INVENTION

[0008] The main purpose of the present invention is to provide an innovative light-sensing device and its manufacturing method to improve the following problems of the conventional technology. First, the manufacturing process is complicated, low yield, slow production, and high cost. Second, the module configuration is fixed and the flexibility of change is small. Third, it is impossible to accurately interpret the signal and the measurement accuracy will be influenced.

[0009] To achieve the above objective, the present invention discloses a light-sensing device which includes a substrate, at least one light-emitting component and a light-sensing component. The light-emitting component is used to emit a light, and the light-sensing component is used to receive the diffuse-reflected light after the light is diffuse-reflected by an external object. Moreover, the light-emitting component and the light-sensing component are individually and separately mounted on the substrate.

[0010] In one embodiment of the light-sensing device of the present invention, the light-sensing device further includes at least one light-blocking component surrounding the at least one light-emitting component and the light-sensing component to block the light emitted by the at least one light-emitting component from leaking out of the light-sensing device and also block an ambient light and the light from being directly received by the light-sensing component without reflection.

[0011] In one embodiment of the light-sensing device of the present invention, the material of the at least one light-blocking component is selected from one of the group consisting of thermoplastic polymers, carbon fiber materials, ceramic materials, and their combinations.

[0012] In one embodiment of the light-sensing device of the present invention, the transmittance rate of the at least one light-blocking component is not greater than 5%.

[0013] In one embodiment of the light-sensing device of the present invention, the reflection rate of the at least one light-blocking component is not less than 95%.

[0014] In one embodiment of the light-sensing device of the present invention, the light-sensing device further includes a pressure-sensitive component disposed on the at least one light-blocking component for sensing an external force exerted on the light-sensing device.

[0015] In one embodiment of the light-sensing device of the present invention, the external force sensed by the pressure-sensitive component and applied to the light-sensing device is less than 100 Newtons (N).

[0016] In one embodiment of the light-sensing device of the present invention, the light-sensing device further includes an optical film plate covering the light-sensing component for selectively allowing only light of a specific wavelength to pass through and blocking light of other wavelengths from passing through.

[0017] To achieve the above objective, the present invention discloses a method of manufacturing a

light-sensing device which includes: providing a substrate, providing at least one light-emitting component, and providing a light-sensing component, and mounting the at least one light-emitting component and the light-sensing component to the substrate individually and separately, wherein the light-sensing component is used to receive a diffuse-reflected light after the light is diffuse-reflected by an external object.

[0018] In one embodiment of the method of manufacturing a light-sensing device of the present invention, the method further includes a step of providing a least one light-blocking component surrounding the at least one light-emitting component and the light-sensing component to block the light emitted by the at least one light-emitting component from leaking out of the light-sensing device and also block an ambient light and the light from being directly received by the light-sensing component without reflection.

[0019] In one embodiment of the method of manufacturing a light-sensing device of the present invention, the method further includes a step of providing a pressure-sensitive component disposed on the at least one light-blocking component for sensing an external force exerted on the light-sensing device.

[0020] In one embodiment of the method of manufacturing a light-sensing device of the present invention, wherein the step of providing at least one light-emitting component and a light-sensing component is the step of: mounting the at least one light-emitting component and the light-sensing component to a conductive substrate, providing a protective layer to cover the at least one light-emitting component and the light-sensing component, and cutting the conductive substrate to form the at least one light-emitting component and the light-sensing component individually and separately.

[0021] In one embodiment of the method of manufacturing a light-sensing device of the present invention, the method further includes a step of providing an optical film cover to cover the protective layer and then cutting the conductive substrate.

[0022] After reviewing the diagrams and the subsequent descriptions of the embodiments, those skilled in the art will readily understand other objectives of the present invention, as well as the technical means and embodiments thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. 1 depicts a schematic cross-sectional view of a conventional near-infrared blood glucose detection device;

[0024] FIG. 2A and FIG. 2B illustrate schematic diagrams of the multi-chip module in the conventional blood glucose detection device due to the user wearing too loosely or too tightly;

[0025] FIG. 3 illustrates a structural schematic diagram of a light-sensing device in one embodiment of the present invention;

[0026] FIG. 4 illustrates a schematic diagram of the combination of light-emitting components, light-sensing components, and light-blocking components in one embodiment of the present invention;

[0027] FIG. 5 depicts a flowchart of manufacturing a light-sensing device in one embodiment of the present invention;

[0028] FIG. 6 illustrates a schematic diagram of manufacturing light-emitting components and light-sensing components in one embodiment of the present invention; and

[0029] FIG. 7 shows a schematic diagram of manufacturing light-emitting components and light-sensing components in another embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0030] The content of the present invention will be explained through examples. The embodiments

of the present invention are not intended to limit the invention to any particular environment, application, or specific method as described in the examples. Therefore, the description of the embodiments is only intended to elucidate the objectives of the present invention and not to limit the invention. It should be noted that in the following embodiments and diagrams, devices not directly related to the present invention have been omitted and not depicted. Additionally, the dimensional relationships between devices in the diagrams are for ease of understanding and are not intended to limit the actual proportions.

[0031] Please refer to FIG. 3, which illustrates a schematic diagram of a light-sensing device in one embodiment of the present invention. One of the differences between the light-sensing device of the present invention and the conventional device is that the present invention utilizes a single-chip module instead of the multi-chip module used in the conventional device. Both the light-emitting component providing the light source and the light-sensing component generating electrical signals from the received light are independently designed as single-chip modules in the structure aiming to enhance the flexibility of module configuration and the possibility of replacing defective modules during manufacturing production. Further details are described below.

[0032] As shown in FIG. 3, the light-sensing device **100** of the present invention comprises a substrate **110**, two light-emitting components **120**, a light-sensing component **130**, and several light-blocking components **140**. Each light-emitting component **120** includes a light-emitting diode (LED) chip **122** and a conductive substrate **124**. The LED chip **122** is mounted on the conductive substrate **124** and electrically connected to it. The LED chip **122** is used to emit light of a specific wavelength. In the specific application of glucose concentration detection, the LED chip **122** can emit near-infrared light with wavelengths ranging from 700 nanometers (nm) to 2500 nm. In other embodiments, different LED chips emitting different wavelengths can be used according to different detection requirements. It should be noted that each light-emitting component **120** further includes a protective layer **126** covering the LED chip **122** to protect it and form a light-emitting package. Additionally, in the embodiment shown in FIG. 3, each light-emitting component **120** further includes an optical film plate **128** covering the protective layer **126** and the LED chip **122**, wherein the optical film plate **128** comprises a glass cover plate **128a** and an optical coating **128b**.

[0033] Furthermore, the light-sensing component **130** comprises a light-sensing chip **132** and a conductive substrate **134**. The light-sensing chip **132** is mounted on the conductive substrate **134** and electrically connected to it. The light emitted by the light-emitting component **120** is diffusely reflected by an external object, such as human tissue. The diffusely reflected light is received by the light-sensing chip **132** of the light-sensing component **130**, which generates a corresponding electrical signal based on it. After conversion by the system, the corresponding blood glucose concentration within, for example, human tissue is obtained. Similarly, the light-sensing component **130** also includes a protective layer **136** covering the light-sensing chip **132** to protect it and form a light-sensing package. Additionally, the light-sensing component **130** includes an optical film plate **138** covering the protective layer **136** and the light-sensing chip **132**, wherein the optical film plate **138** comprises a glass cover plate **138a** and an optical coating **138b**. It should be noted that this optical film plate **138** can be an optical filter, such as a bandpass filter (BPF), used to selectively allow only light within a specific wavelength range to pass through while blocking or attenuating light of other wavelengths.

[0034] Moreover, unlike the prior art, the light-emitting component **120** and the light-sensing component **130** in the present invention are separate and distinct components. They are individually and separately installed on a substrate **110** with printed circuits during the subsequent production process of the light-sensing device **100**. Therefore, different numbers of light-emitting components **120** and light-sensing components **130** can be installed according to the different final requirements of the light-sensing device **100**, providing flexibility in product configuration. Furthermore, even when one of the components is defective, the light-sensing device **100** of the present invention still offers the possibility of replacing the faulty components on the substrate **110**. Detailed explanations

regarding this aspect can be provided in the subsequent description of the manufacturing method for the light-sensing device **100** of the present invention.

[0035] Continuing with reference to FIG. **3**, another difference between the present invention and the prior art lies in the inclusion of at least one light-blocking component **140** in the present invention's light-sensing device **100**. This component surrounds the light-emitting component **120** and the light-sensing component **130** to prevent light emitted by the light-emitting component **120** from leaking outside the light-sensing device **100**. It also blocks ambient light and prevents light that has not been reflected by external objects (such as human tissue) from being directly received by the light-sensing component **130**. Specifically, please refer to FIG. **4**, which shows that the shape of the light-blocking component can be designed in different forms according to requirements, such as, but not limited to, an “O” shaped or a “C” shaped structure. In the embodiment shown in FIG. **4**, an “O” shaped light-blocking component **140** and two “C” shaped light-blocking components **140** are used to enclose two light-emitting components **120** and one light-sensing component **130** entirely for achieving the aforementioned purpose of preventing light leakage and blocking ambient light and specific wavelength light from entering the light-sensing component **130**. The height of the light-blocking component **140** can be, but is not limited to, greater than or equal to 0.4 millimeters (mm), and its width is not less than 0.1 millimeters (mm). Furthermore, the material composing the light-blocking component **140** can be selected from one or a combination of thermoplastic polymers, carbon fiber materials, ceramic materials, among which the thermoplastic polymers may include polystyrene, low-density polyethylene, polyacetal, polylactic acid, acrylonitrile-butadiene-styrene copolymer, etc. Additionally, the transmittance rate of the material constituting the light-blocking component should not exceed 5%, or the reflection rate should not be less than 95% to achieve the effectiveness of blocking light.

[0036] In a preferred embodiment of the present invention, to avoid problems arising from differences in user operation status during the actual application of the light-sensing device and the occurrence of loose or tight wearing conditions between the user's body tissue and the device, the light-sensing device **100** of the present invention further includes a pressure-sensitive component **150**. This component can be adhered to the contour above the light-blocking component **140** and is placed on the light-blocking component **140** to sense the external force applied to the light-sensing device **100**. Specifically, when the light-sensing device **100** is worn too loosely against the body tissue, the pressure-sensitive component **150** may sense zero external force applied to the light-sensing device **100** or approach zero. At this point, the light-sensing device **100** can emit a warning signal indicating loose wearing, for reminding the user to readjust the wearing state. On the other hand, when the light-sensing device **100** is worn too tightly against the body tissue causing the pressure-sensitive component **150** to sense an external force exceeding the preset value of the system, for example, it could be 100 Newtons (N), the light-sensing device **100** can emit a corresponding warning indicating tight wearing for reminding the user to readjust the wearing state to obtain accurate measurement values, such as blood glucose concentration.

[0037] Please refer to FIG. **5**, which illustrates a flowchart of the process for manufacturing the light-sensing device according to the present invention. In step **501**, a substrate containing printed circuits is provided. In step **502**, at least one light-emitting component is provided according to the requirements of the light-sensing device to emit light in a specific wavelength range. In step **503**, a light-sensing component is provided according to the requirements of the light-sensing device to receive the diffuse light reflected by external objects from the light emitted by the light-emitting component. In step **504**, a specific number of individual and separate light-emitting components and light-sensing components are installed on the substrate according to the requirements of the light-sensing device. In step **505**, a specific number of light-blocking components are provided according to the requirements of the light-sensing device to surround the above-mentioned light-emitting components and light-sensing components for blocking the light emitted by the light-emitting components from leaking outside the light-sensing device, and also blocking ambient light

and the light not reflected by external objects from being directly received by the light-sensing components. In step **506**, a pressure-sensitive component is provided and placed on the light-blocking component to sense the external force applied to the light-sensing device.

[0038] The following will describe how the present invention manufactures individual and separate light-emitting components and light-sensing components in a light-sensing device of two embodiments. Please refer to FIG. **6**. In Process **6A**, several light-emitting diode chips **602** and several light-sensing chips **604** fixed on a conductive substrate **606** by die bonding or surface adhesive technology are shown. Next, in Process **6B**, wire bonding is performed to electrically connect the light-emitting diode chips **602** and the light-sensing chips **604** to the conductive substrate **606**. Subsequently, in Process **6C**, a protective layer **608** is formed by a molding process to cover the light-emitting diode chips **602** and the light-sensing chips **604**. Then, in Process **6D**, an optical thin film cover plate **610** with optical thin films is provided to cover the protective layer **608**, simultaneously covering the light-emitting diode chips **602** and the light-sensing chips **604**. Next, in Process **6E**, the conductive substrate **606** is cut using cutting equipment to form individual and separated, fully encapsulated light-emitting components **612** and light-sensing components **614**. Then, in Process **6F**, according to the design requirements of the light-sensing device, a certain number of individual and separated light-emitting components **612** and light-sensing components **614** are assembled onto a printed circuit board **616**. In Process **6G**, a suitable number and structural design of light-blocking components **618** are used to cover the entire periphery of the light-emitting components **612** and light-sensing components **614**. Finally, in Process **6H**, a pressure-sensitive component **620** is adhered to the assembled light-blocking component **618** so the light-sensing device **600** of the present invention has been done.

[0039] Please refer to FIG. **7**, which illustrates another embodiment of how the present invention manufactures individual and separate light-emitting components and light-sensing components. First, similarly, in Process **7A**, several light-emitting diode chips **702** and several light-sensing chips **704** are fixed onto a conductive substrate **706** using die bonding or surface adhesive technology. However, unlike the previous embodiment, the surface of the light-sensing chips **704** in this embodiment is already attached with an optical thin film, which can filter out light in a specific wavelength range from being received by the light-sensing chips **704**. Therefore, the subsequent process of attaching an optical thin film cover plate can be saved. Next, in Process **7B**, wire bonding is performed to electrically connect the light-emitting diode chips **702** and the light-sensing chips **704** to the conductive substrate **706**. Subsequently, in Process **7C**, a protective layer **708** is formed by a molding process to cover the light-emitting diode chips **702** and the light-sensing chips **704**. Then, in Process **7D**, the conductive substrate **706** is cut using cutting equipment to form individual and separate, fully encapsulated light-emitting components **712** and light-sensing components **714**. Next, in Process **7E**, according to the design requirements of the light-sensing device, a certain number of individual and separate light-emitting components **712** and light-sensing components **714** are assembled onto a printed circuit board **716**. In Process **7F**, a light-blocking component **718** is used to cover the entire periphery of the light-emitting components **712** and light-sensing components **714**. Finally, in Process **7G**, a pressure-sensitive component **720** is adhered to the assembled light-blocking component **718** to complete the light-sensing device **700** of the present invention.

[0040] As mentioned above, the application of “single-chip module” pattern for light-emitting components and light-sensing components in the present invention allows for the selection of different numbers of module components according to the requirements of different applications, and with different structural shapes, such as modular light-blocking components in “O” shaped or “C” shaped designs. These components can be freely combined on a printed circuit board for providing flexibility in diverse combinations without limitations on the quantity, position, or arrangement thereof. Moreover, if any module is defective, the light-sensing device of the present invention offers the possibility of replacing it with a good one. It is not necessary to discard the

entire module as the traditional manner. Therefore, the light-sensing device of the present invention can address the drawbacks of traditional multi-chip module glucose monitoring devices. The modified design can: (1) improve light-blocking efficiency due to a complete light-blocking component design, (2) enhance module yield by replacing defective products, (3) increase production speed with single-chip module design, (4) reduce overall device production costs with single-chip module design flexibility, and (5) provide diverse choices to meet user needs with increased module matching flexibility. Particularly, the light-sensing device of the present invention features a pressure-sensitive component that provides force sensing functionality for enabling the non-invasive glucose monitoring module of the present invention to screen or calibrate measurement data. Thereby, glucose monitoring accuracy will be enhanced.

[0041] The above embodiments are essentially for illustrative purposes only and are not intended to limit the embodiments of the claims or the applications or uses thereof. Furthermore, although at least one exemplary embodiment has been presented in the foregoing embodiments, it should be understood that the present invention may have numerous variations. It should also be understood that the embodiments described herein are not intended to limit the scope, applications, or configurations of the claimed subject matter in any way. On the contrary, the foregoing embodiments may provide a convenient guide for those skilled in the art to implement one or more embodiments disclosed herein. Additionally, various changes may be made to the functions and arrangements of the components without departing from the scope defined by the claims, and the claims encompass all known equivalents and foreseeable equivalents at the time of filing of the present patent application.

Claims

1. A light-sensing device, including: a substrate; at least one light-emitting component, used to emit a light; and a light-sensing component, used to receive a diffuse-reflected light after the light is diffuse-reflected by an external object, wherein the light-emitting component and the light-sensing component are individually and separately mounted on the substrate.
2. The light-sensing device of claim 1, further including at least one light-blocking component surrounding the at least one light-emitting component and the light-sensing component to block the light emitted by the at least one light-emitting component from leaking out of the light-sensing device and also block an ambient light and the light from being directly received by the light-sensing component without reflection.
3. The light-sensing device of claim 2, wherein the material of the at least one light-blocking component is selected from one of the group consisting of thermoplastic polymers, carbon fiber materials, ceramic materials, and their combinations.
4. The light-sensing device of claim 2, wherein the transmittance rate of the at least one light-blocking component is not greater than 5%.
5. The light-sensing device of claim 2, wherein the reflection rate of the at least one light-blocking component is not less than 95%.
6. The light-sensing device of claim 2, further including a pressure-sensitive component disposed on the at least one light-blocking component for sensing an external force exerted on the light-sensing device.
7. The light-sensing device of claim 6, wherein the external force sensed by the pressure-sensitive component and applied to the light-sensing device is less than 100 Newtons (N).
8. The light-sensing device of claim 1, further including an optical film plate covering the light-sensing component for selectively allowing only light of a specific wavelength to pass through and blocking light of other wavelengths from passing through.
9. A method of manufacturing a light-sensing device, including: providing a substrate; providing at least one light-emitting component; and providing a light-sensing component, and mounting the at

least one light-emitting component and the light-sensing component to the substrate individually and separately, wherein the light-sensing component is used to receive a diffuse-reflected light after the light is diffuse-reflected by an external object.

10. The method of manufacturing a light-sensing device of claim 9, further including a step of providing a least one light-blocking component surrounding the at least one light-emitting component and the light-sensing component to block the light emitted by the at least one light-emitting component from leaking out of the light-sensing device and also block an ambient light and the light from being directly received by the light-sensing component without reflection.

11. The method of manufacturing a light-sensing device of claim 10, further including a step of providing a pressure-sensitive component disposed on the at least one light-blocking component for sensing an external force exerted on the light-sensing device.

12. The method of manufacturing a light-sensing device of claim 9, wherein the step of providing at least one light-emitting component and a light-sensing component is the step of: mounting the at least one light-emitting component and the light-sensing component to a conductive substrate, providing a protective layer to cover the at least one light-emitting component and the light-sensing component, and cutting the conductive substrate to form the at least one light-emitting component and the light-sensing component individually and separately.

13. The method of manufacturing a light-sensing device of claim 12, further including a step of providing an optical film cover to cover the protective layer and then cutting the conductive substrate.
